CNNs: Tackling Overfitting for Better Generalization

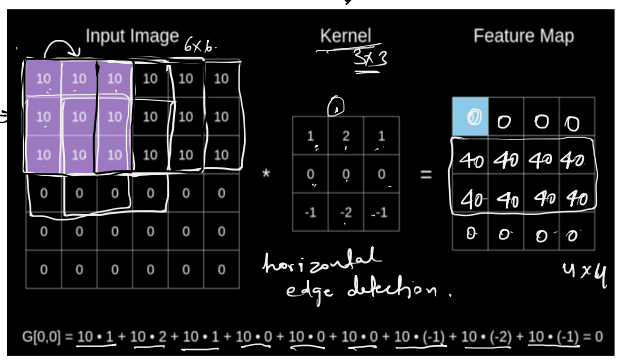
Saturday, 31 October 2026 8:03 PM

Last class → Trained a ANN/MLP on image data → ≈ 50% Train Accuracy, ≈ 50% Val Accuracy. } Did not work well.

Trained a CNN model on image data → ≈ 90% Train Accuracy, ≈ 50% Val Accuracy } Worked well but we had overfitting.

Convolution Operation.

(kernel / filter 3x3)



Input 6x6

Element wise multiplication

Initial layers. → | — [7 ( )

Input image →





← Vertical edges

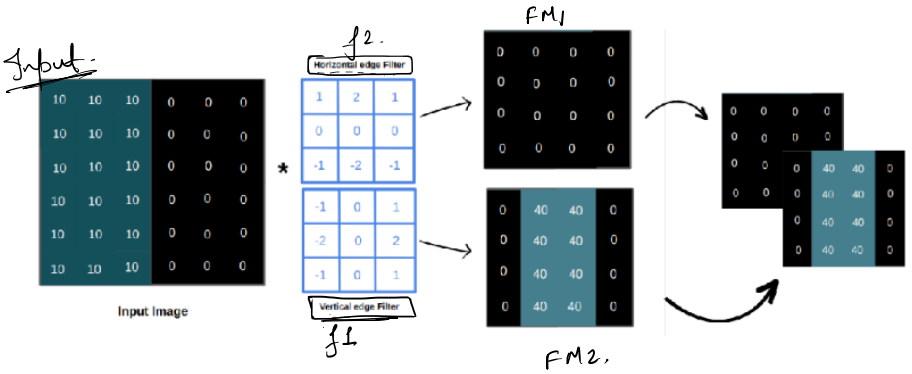
Sobel filter.

-1	0	1
-2	0	2
-1	0	1

← Vertical edge detection.

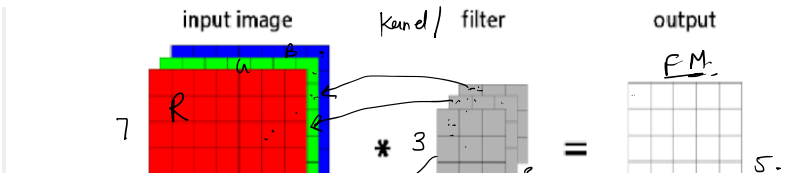
Vertical edge → |

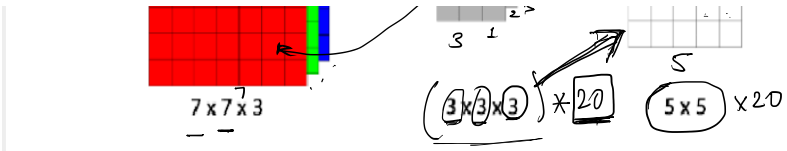
horizontal → —



5x5, 5x5, 5x5 } Element wise Sum

5x5





Color Image → R G B

Q. What is the output shape for Input Image (7x7), Kernel (3x3). Stride = 1 and Padding = 0?

$$output = \frac{input - kernel_size + 2 * padding}{stride} + 1$$

Stride → how many steps does my kernel shift?

$$output = \frac{7 - 3 + (2*0)}{1} + 1$$

$$(4 + 0) + 1 = 5 \times 5$$

Image → 6x6

Kernel → 3x3

Stride → 1

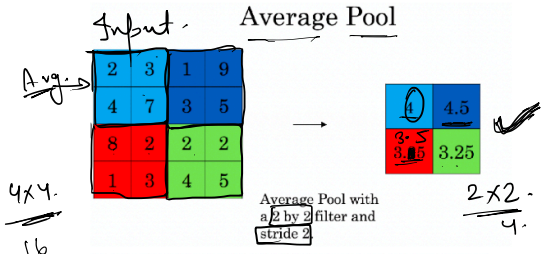
Padding → 0

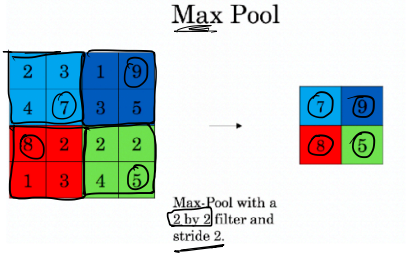
$$\frac{6 - 3 + 2*0}{1} + 1$$

$$\frac{3 + 0}{1} + 1 = 4 \times 4$$

* Pooling:

50x50 → Max → 4x4 → 16



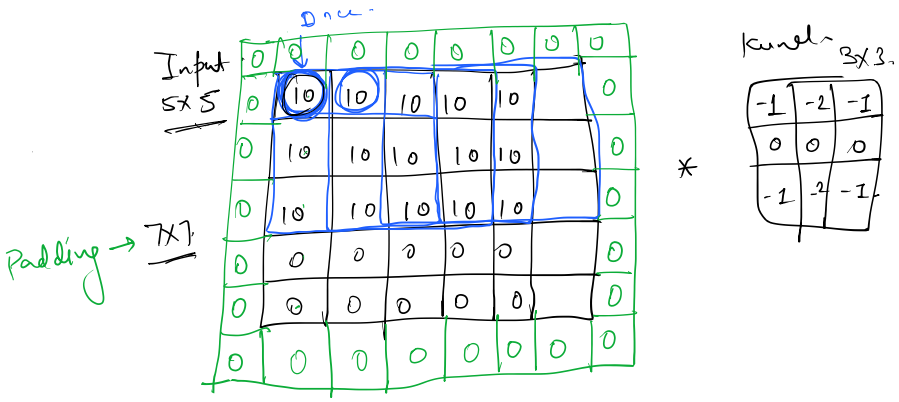


* Flattening





* Padding.



$I_{5\times5} * K_{3\times3} \rightarrow$

$$\frac{5 - 3 + 2\times0}{1} + 1 \rightarrow 3\times3 \text{ (output)}$$

$$5 - 3 + 2\times1 + 1 \rightarrow 5\times5.$$

